

Towhid Ahmed

Agentic AI Engineer · Open to remote — worldwide

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Professional Summary

Agentic AI Engineer with **1+ year of freelance experience** building **AI and Generative AI applications** using **Python, Machine Learning, FastAPI, LLMs, RAG, LangChain, and LangGraph**. Built **3 production-oriented AI applications**, improving **retrieval relevance by 25%**, reducing **hallucinations by 35%**, and achieving **98/98 passing automated tests**. Experienced in **RAG pipelines, LLM Ops, Docker, and LangSmith**.

Technical Skills

Agentic AI: AI Agents, Multi-Agent Systems, Agent Orchestration, Tool Calling, Agent Routing, Agent Memory

Frameworks: LangGraph, LangChain

LLMs & Models: LLMs, Prompt Engineering, OpenAI, Llama, Groq, Hugging Face

RAG & Retrieval: RAG, Hybrid Retrieval, Semantic Search, Reranking, Embeddings, Citation-Based RAG

Databases: Pinecone, ChromaDB, PostgreSQL

Backend & APIs: Python, FastAPI, REST APIs, Async Programming, Docker, Git, GitHub Actions, CI/CD

LLM Ops & Testing: LangSmith, LLM Evaluation, LLM Guardrails, Pytest

Professional Experience

Freelance Agentic AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | Jun 2026 – Jul 2026

- Enhanced information reliability and user trust by developing a **3-agent LangGraph workflow** with **citation-based RAG**, significantly **reducing AI hallucinations**.
- Powered seamless document processing and interaction by engineering **7 production-ready REST APIs** using **FastAPI, OpenAI API, and Pinecone**.
- Enabled **1,000+** document knowledge-base updates without redeployment.

AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025

- **Automated customer interactions and improved consistency** using a multi-agent **LangGraph pipeline** (Intent, Retrieval, Response) with Pinecone RAG and **conversational memory**.
- Slashed AI inference costs and maintained high execution speed by integrating **Groq**, Hugging Face embeddings, and **Pinecone Serverless**.
- **Built a scalable support infrastructure** with a Streamlit UI and a FastAPI backend featuring **6+ REST APIs** for knowledge uploads and session management.

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, Hugging Face, Tavily GitHub | Demo

- **Designed 4-agent LangGraph workflow** routing queries between **RAG** and **web search** based on intent and complexity.
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**.
- Integrated **Tavily** with **DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**.

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25%** and **reducing hallucinations by 35%**.
- Implemented two-stage retrieval (**semantic chunking** + cross-encoder **re-ranking**) with **<2s latency** via **caching**.
- Tech: **Groq Llama 3.3, BAAI embeddings, Pinecone, FastAPI, Streamlit, Docker**; 76-unit test coverage.

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Built a clinical information assistant using a **4-agent LangGraph pipeline** (Router, Safety, Retrieval, Answer) with **RAG and web search**.
- Implemented **multi-layer safety system** (input filters, retrieval validation, output sanitization) with JWT auth.
- Achieved **98/98** passing tests across **guardrails, RAG evaluation, and API workflows**.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology | 2023-2027

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLM Ops — Udemy • Data Structures & Algorithms — TakeUForward